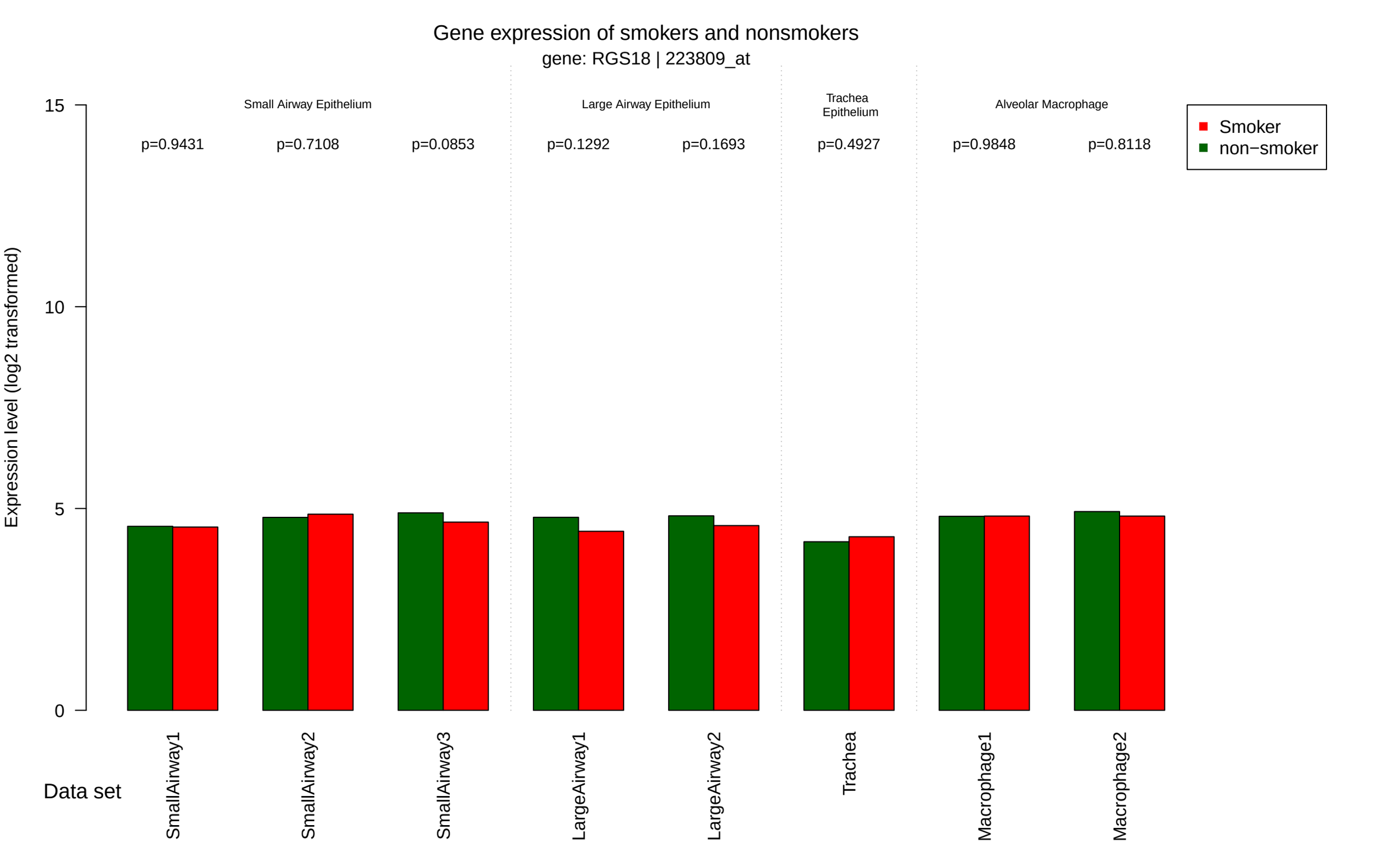
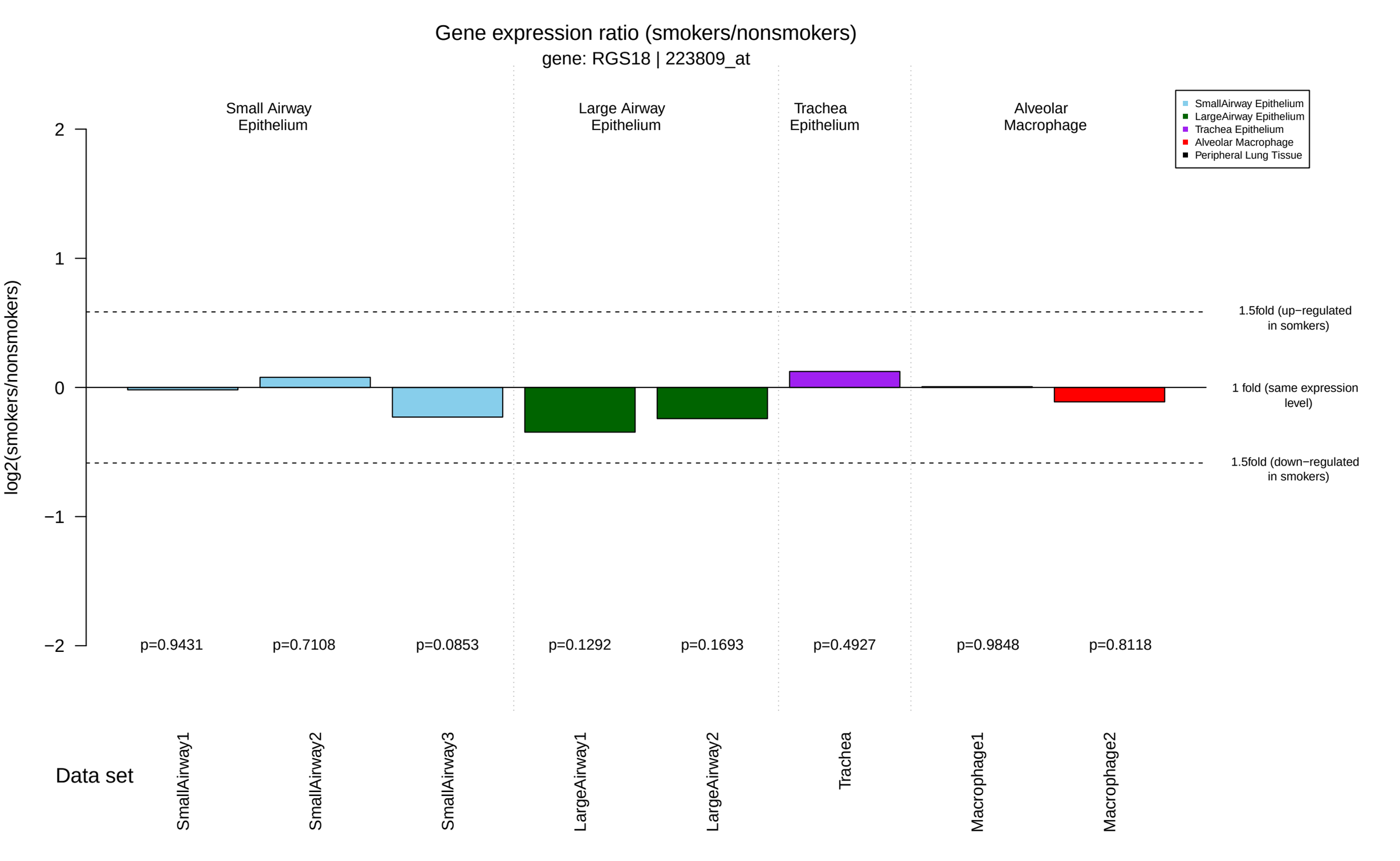
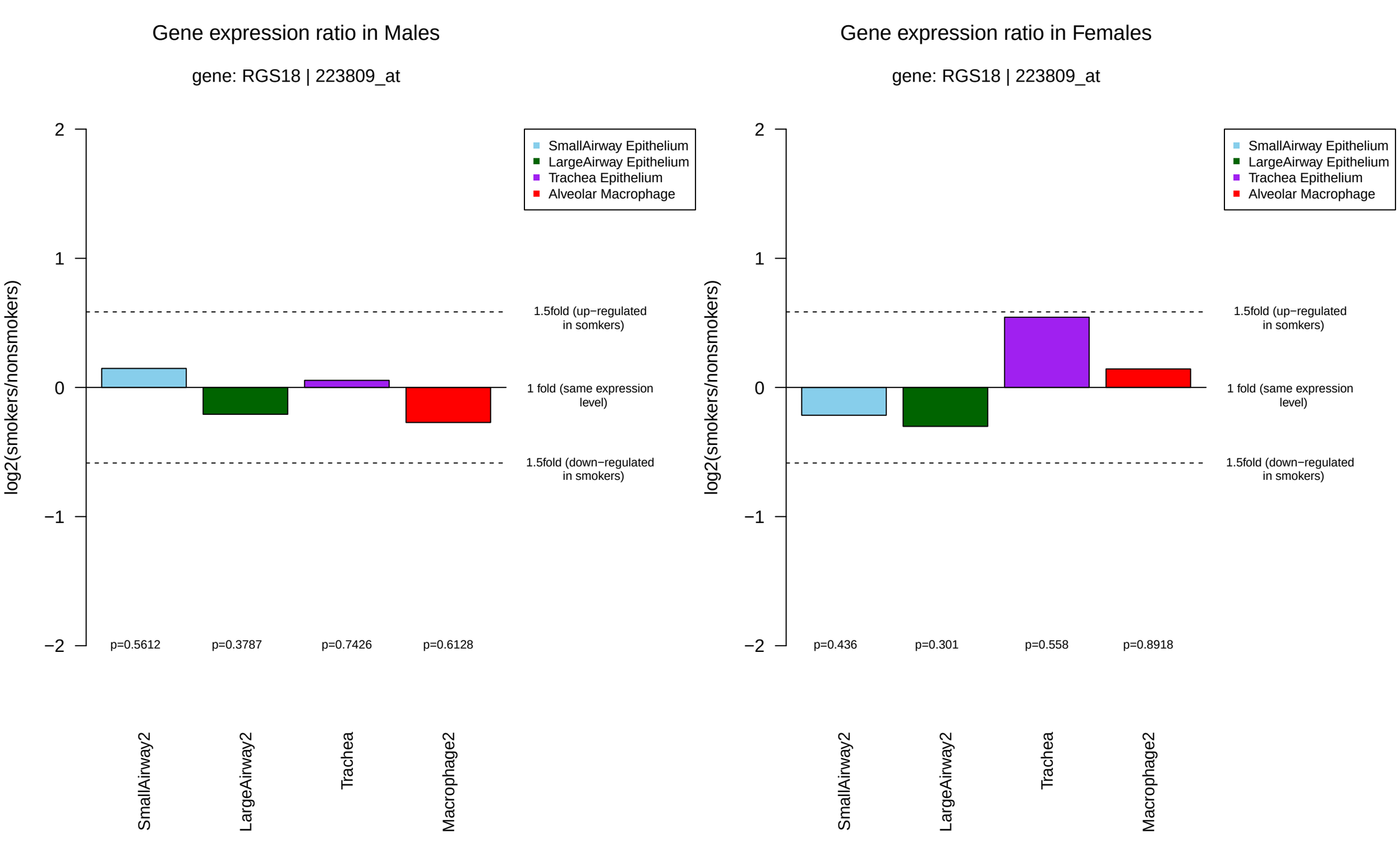
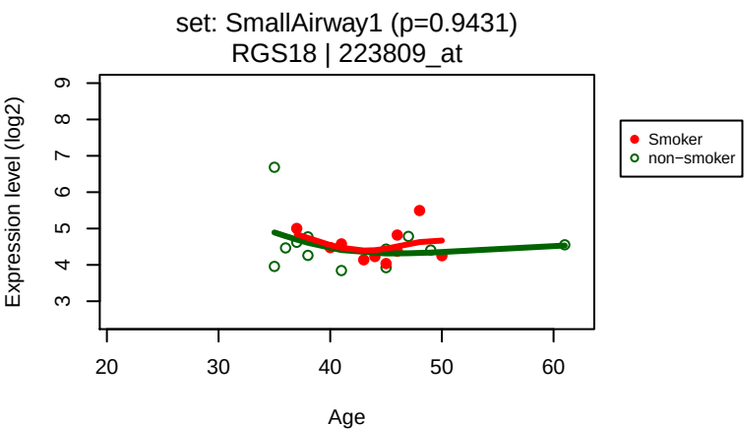
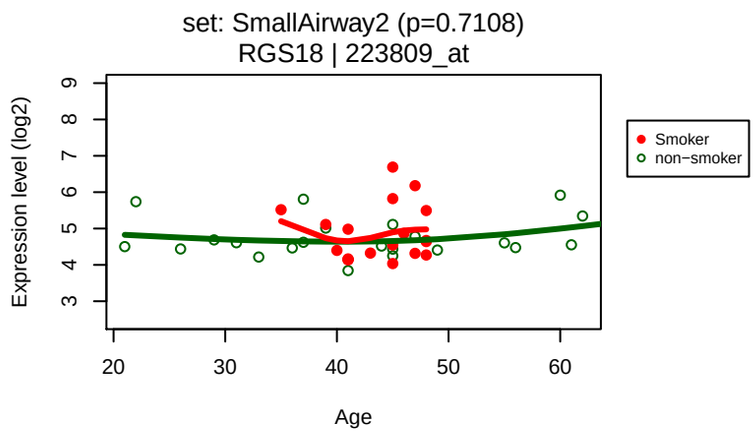
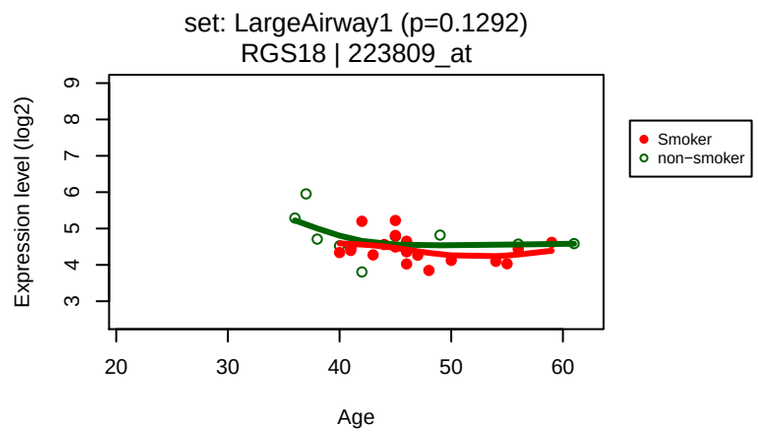

Gene expression of smokers and nonsmokers
gene: RGS18 | 223809_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9431
p=0.7108
p=0.0853
p=0.1292
p=0.1693
p=0.4927
p=0.9848
p=0.8118
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

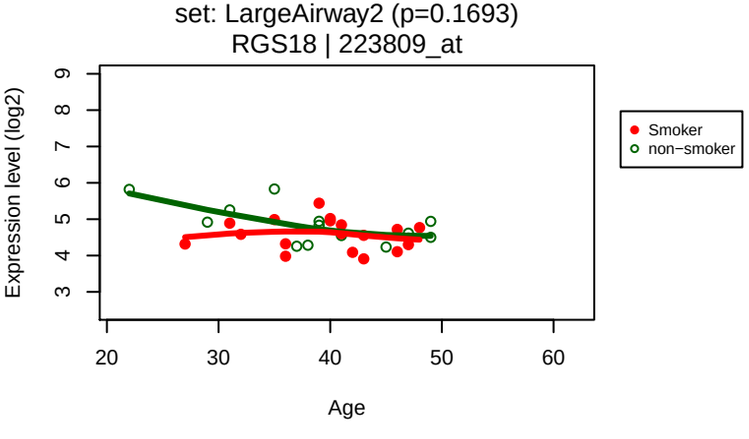

Gene expression ratio (smokers/nonsmokers)
gene: RGS18 | 223809_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9431
p=0.7108
p=0.0853
p=0.1292
p=0.1693
p=0.4927
p=0.9848
p=0.8118
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

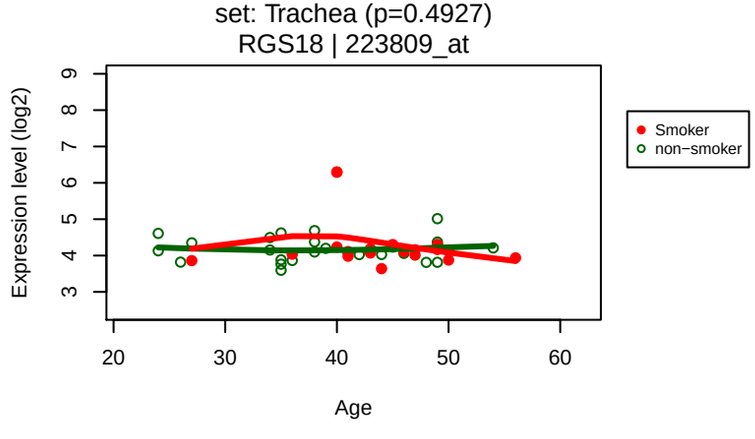

Gene expression ratio in Males
gene: RGS18 | 223809_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5612
p=0.3787
p=0.7426
p=0.6128
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS18 | 223809_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.436
p=0.301
p=0.558
p=0.8918
SmallAirway2
LargeAirway2
Trachea
Macrophage2

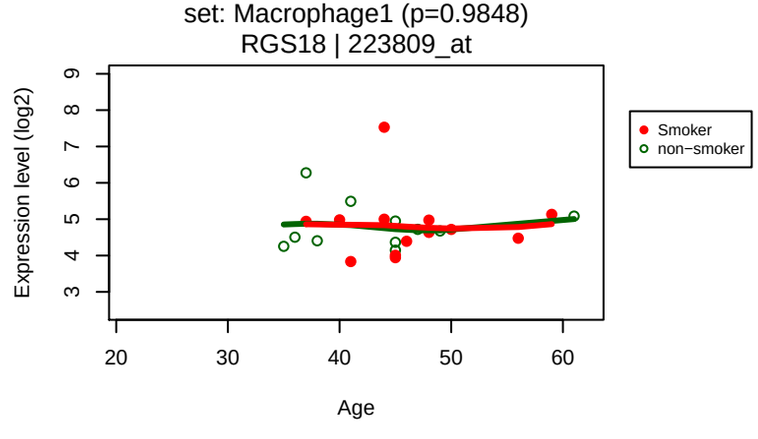

set: SmallAirway1 (p=0.9431)
RGS18 | 223809_at
Expression level (log2)
Age
Smoker
non−smoker

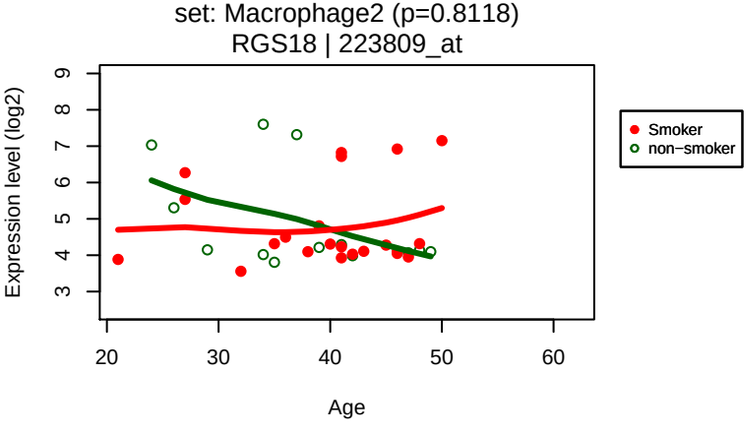

set: SmallAirway2 (p=0.7108)
RGS18 | 223809_at
Expression level (log2)
Age
Smoker
non−smoker

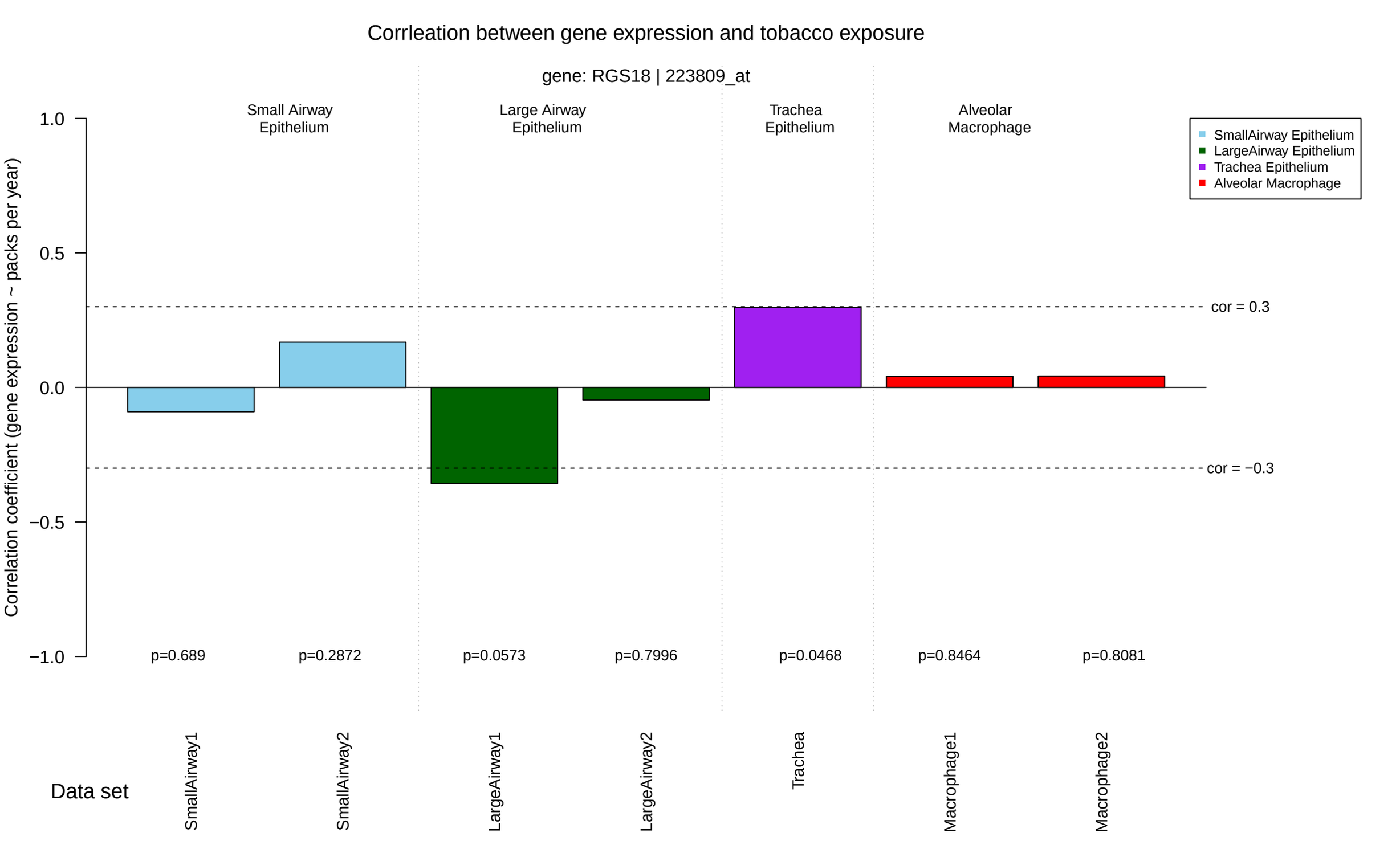

set: LargeAirway1 (p=0.1292)
RGS18 | 223809_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1693)
RGS18 | 223809_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4927)
RGS18 | 223809_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9848)
RGS18 | 223809_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8118)
RGS18 | 223809_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS18 | 223809_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.689
p=0.2872
p=0.0573
p=0.7996
p=0.0468
p=0.8464
p=0.8081
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2